

German Electricity Demand Forecasting Using Statistical, Machine Learning and Deep Learning Models

Link to Repository: <https://github.com/Pavan240571/German-Electricity-Demand-Forecasting>

1. Introduction

Electrical load forecasting is crucial for contemporary power generation systems since there should be a constant balance between electrical generation and consumption. Proper forecasting will allow energy suppliers and system operators to optimize the process of power generation, lower costs, and improve the integration of renewable energy sources. Nevertheless, it is necessary to remember that there are many influencing factors that affect electricity demand, among which seasonal variations, weather changes, and behavioural changes of consumers can be highlighted.

In this research, data is collected from the Open Power System Data (OPSD). The electricity demand for Germany, i.e., DE_load_actual_entsoe_transparency, is used as the variable since the data represents the actual electricity consumption that was collected using the ENTSO-E transparency system. The dataset consists of observations per hour from 1st January 2015 to 31st October 2020. To further analyze the dataset, the Berlin temperature dataset was collected using Open-Meteo historical weather API due to the impact of temperature on electricity consumption.

The purpose of this project is to create various models for predicting German electricity consumption demands. The analysis looks at the conventional statistical methods like benchmark models, SARIMA and SARIMAX along with the machine learning and deep learning models including random forest and gradient boosting and LSTM. These models' effectiveness is measured through forecasting error measurement criteria like MAE and RMSE. The report further offers a critical analysis of these models' performances considering factors such as seasonality, external variable, and model weaknesses, among others.

2. Exploratory Data Analysis (EDA)

Before creating the forecasting models, the dataset for the German electricity demand was investigated in order to gain information about the characteristics of the data. The data spans from January 2015 until October 2020, and it is aggregated to a daily and weekly time series. Summary statistics have been used in order to explore the variations in the data.

There is a definite trend and seasonal effect in the data on the weekly demand for electricity. Demand for electricity tends to be relatively high during winter months but low during summer, implying that there is an annual seasonality in the data.

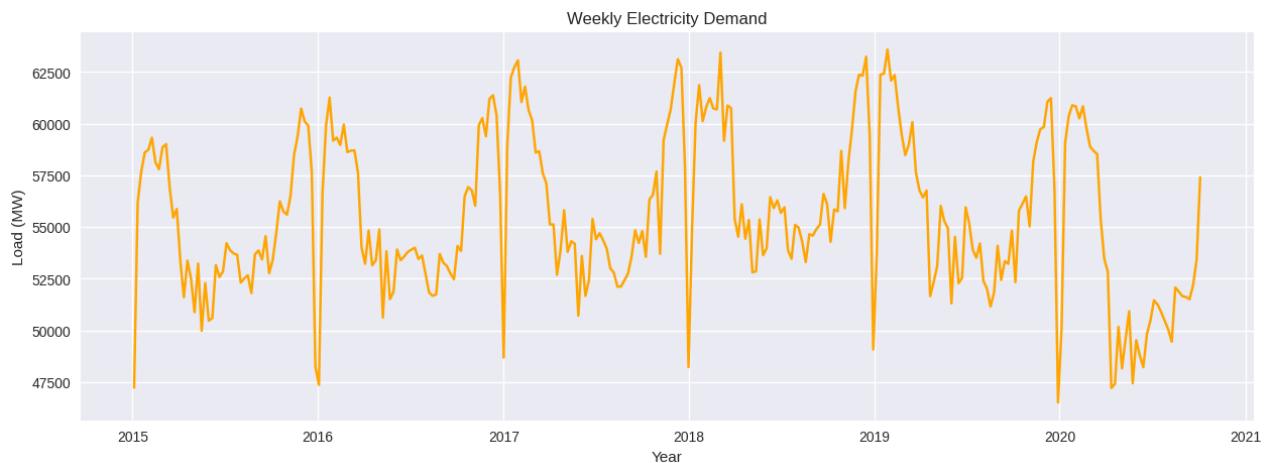


Figure 1. Weekly Electricity Demand Time Series 2015 – 2020

Stationarity was tested through the Augmented Dickey-Fuller test. The output of the test showed that (stationary / non-stationary - insert your findings), which validated the pre-processing done prior to modelling. Seasonal decomposition was used to confirm the presence of trend, seasonality, and residual effects, while ACF and PACF analysis helped show the presence of autocorrelation and seasonal effects.

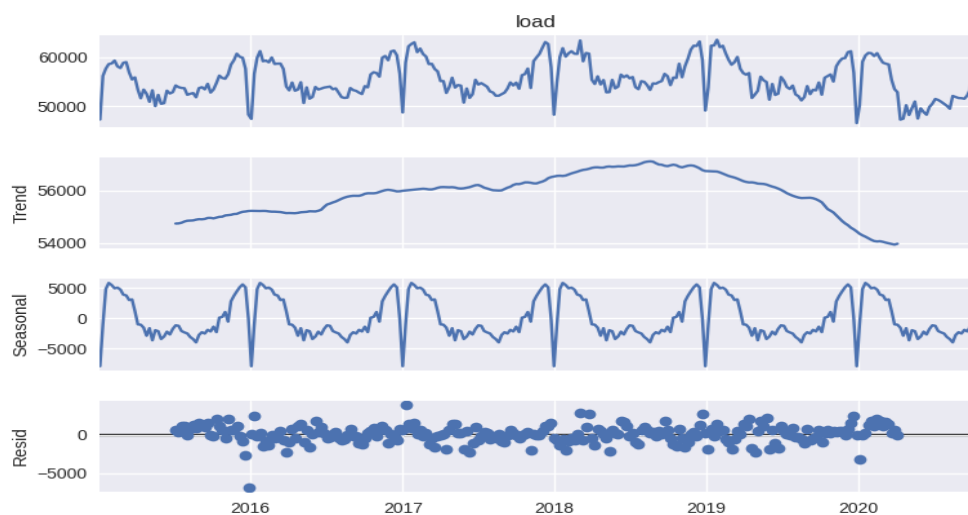


Figure 2. Seasonal Decomposition of Weekly Electricity Demand

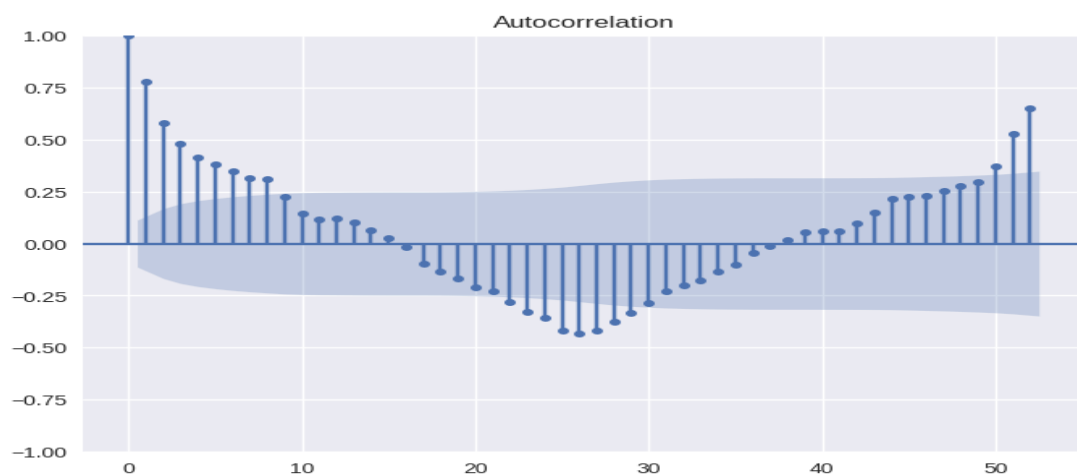


Figure 3. Autocorrelation Function (ACF)

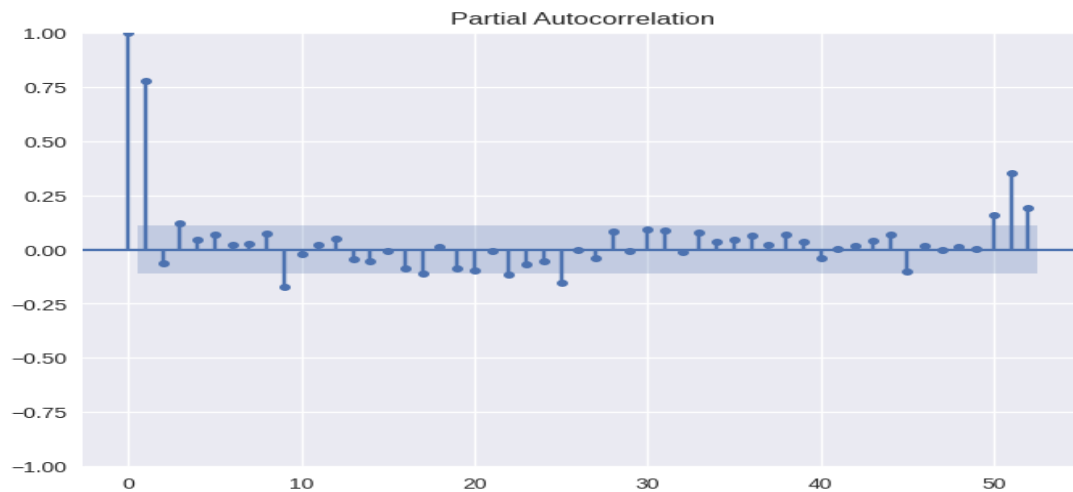


Figure 4. Partial Autocorrelation Function (PACF)

3. Data Preprocessing

The dataset on German electricity demand was prepared through selection of the needed variable, formatting of the timestamp to datetime type, and removal of any missing data. The time range in the dataset was set to run from 1 January 2015 to 31 October 2020.

Original hourly data were converted to daily and weekly averages for the purpose of eliminating fluctuations in the short term in order to capture long-term trends and seasonality. No other transformations, like logarithmic transformation, were used. Differencing was done in SARIMA and SARIMAX when necessary to make the data stationary.

In the data splitting process, 80% of the data is used for training while the remaining 20% is set aside for testing purposes. The chronological order of observations is preserved in the process to avoid data leakage.

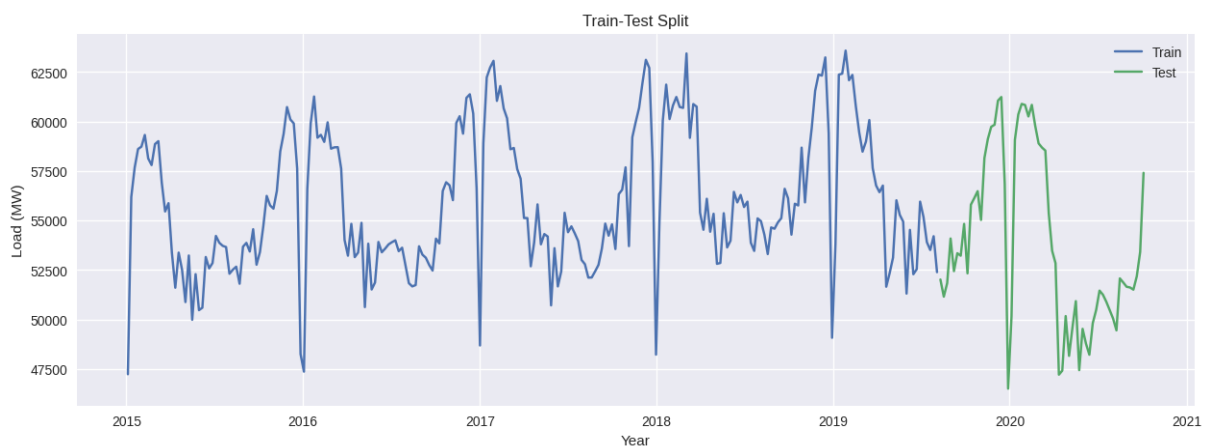


Figure 5. Train-Test Split of Weekly Electricity Demand

The forecasting period was determined based on the test data set, where each model would forecast future electricity consumption on the last part of the time series. The performance of each model was evaluated through the comparison between its predictions and the actual values of electricity consumption.

4. Model Selection and Specification

There were various forecasting models that have been used to forecast electricity consumption in Germany. Some benchmarking models such as the mean, naïve, seasonal naïve, and drift models were initially established.

After identifying the seasonal trends from exploratory analysis, SARIMA was chosen as the method for modelling trend and seasonality. Similarly, the effect of weather on the consumption of electricity was explored using the SARIMAX model in which the Berlin temperature was taken as an external input variable. Furthermore, Random Forest, Gradient Boosting, and LSTM models were used for modelling nonlinear behaviour.

The best values of the SARIMA and SARIMAX parameters were found based on the grid search with the help of the Akaike Information Criterion (AIC). The Random Forest and Gradient Boosting algorithms used predetermined hyperparameters, whereas the LSTM neural network included two LSTM layers and had dropout for regularization purposes.

Assumptions on the model were evaluated using residual diagnostic and training history plots. The residual plot revealed that the SARIMA models managed to capture most of the temporal pattern, while LSTM training and validation loss curves were used to check for model convergence and overfitting.

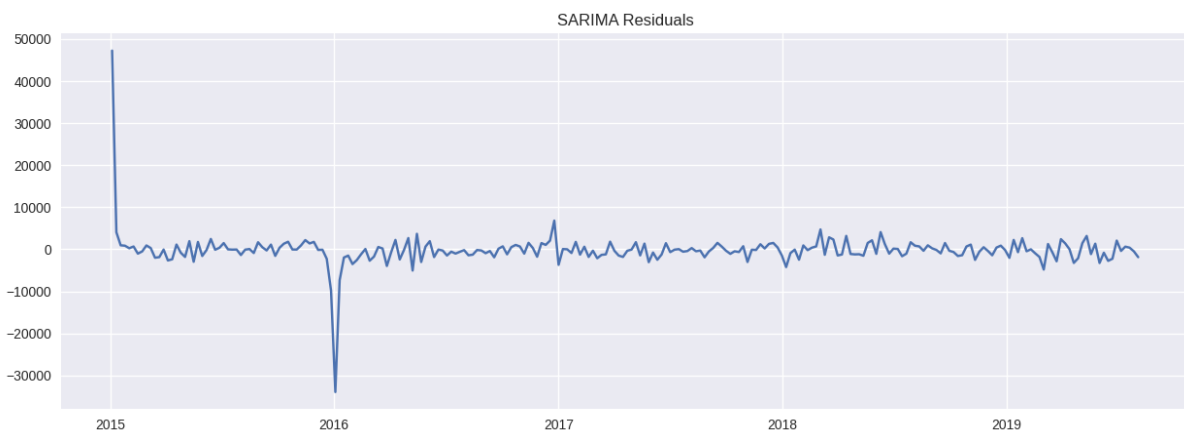


Figure 6. SARIMA Residual Diagnostics (or Residual ACF)



Figure 7. LSTM Training and Validation Loss

5. Model Results and Forecasts

The models were estimated using the training dataset and validated using the test dataset. First, benchmark models were applied as the baseline models, and then the SARIMA, SARIMAX, Random Forest, Gradient Boosting, and LSTM models were employed. The best SARIMA and SARIMAX models were determined based on the lowest values of AIC in the grid search process.

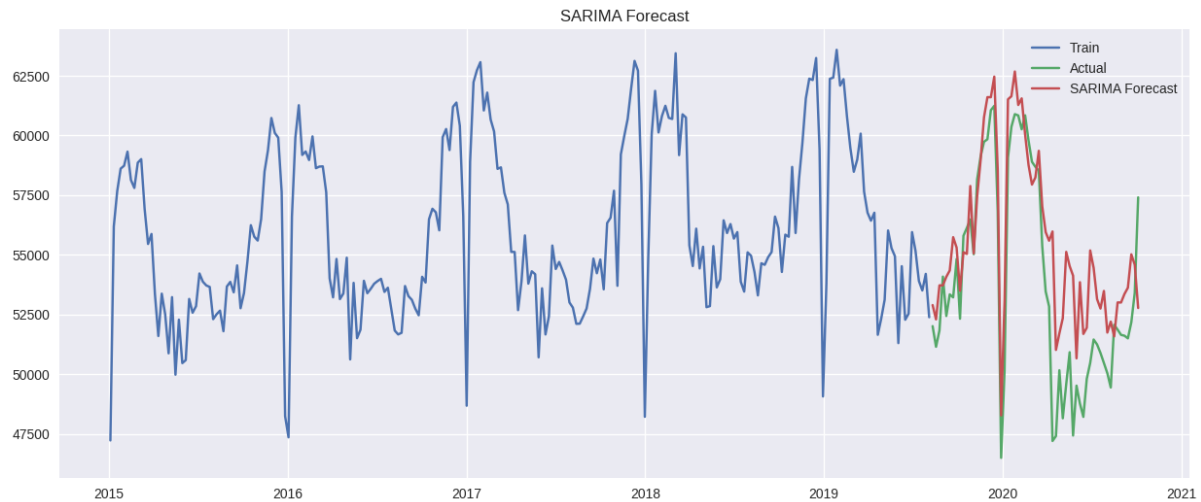


Figure 8. SARIMA Forecast vs Actual Electricity Demand

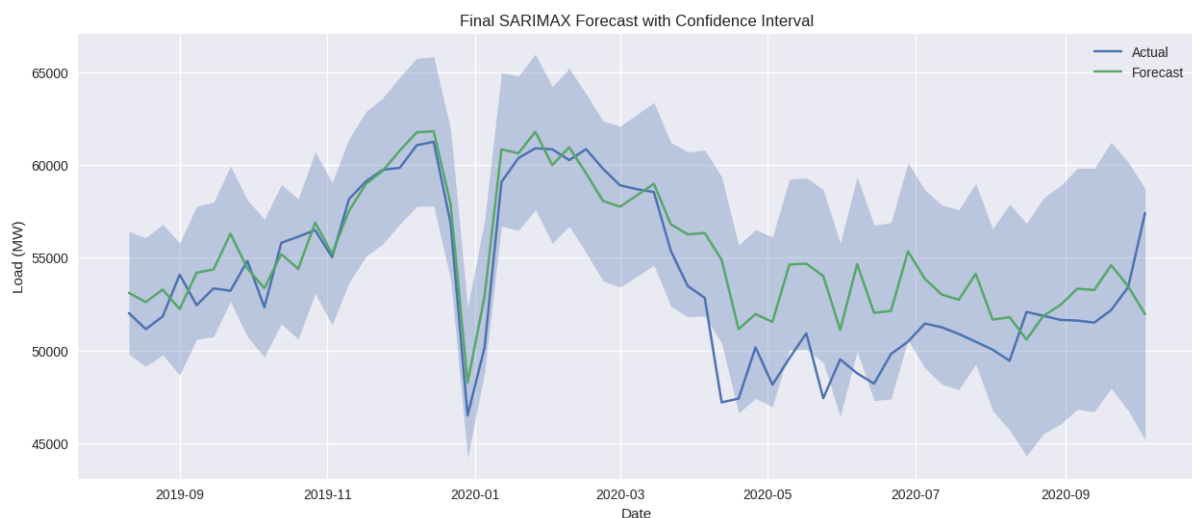


Figure 9. SARIMAX Forecast with Confidence Intervals

As can be seen from the forecasting graphs, both models SARIMA and SARIMAX have been successful in capturing the general trend as well as the seasonality of electricity consumption. The inclusion of temperature as an exogenous variable in the SARIMAX model has helped in capturing the fluctuations in demand.

The predictions from LSTM model also showed strong alignment with electricity demand observations, highlighting its ability to learn complex time series patterns. Random Forest and Gradient Boosting machine learning models were able to provide competitive forecasting accuracy using feature engineering and non-linear modelling techniques.

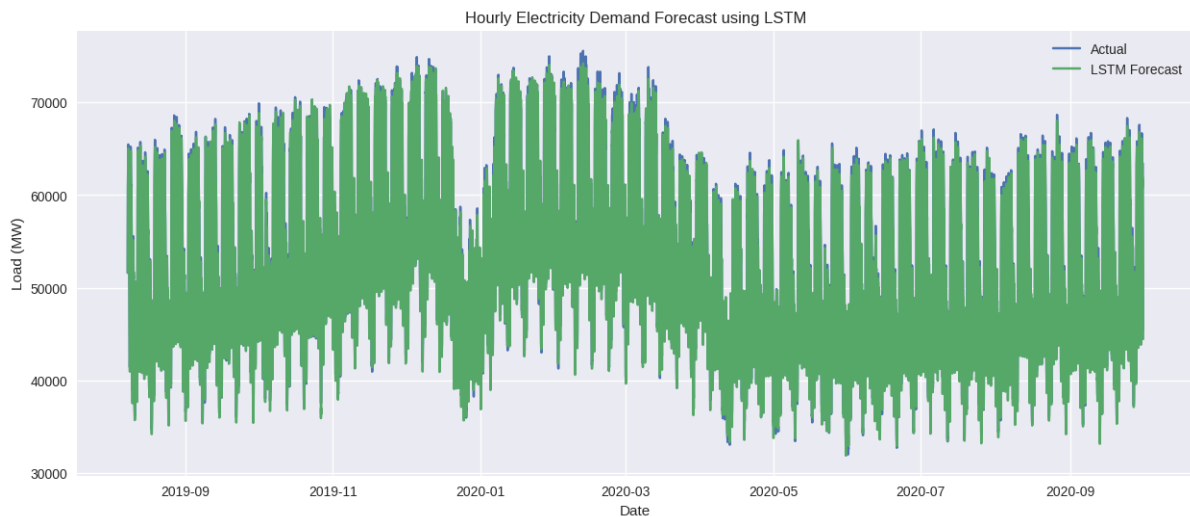


Figure 10. LSTM Forecast vs Actual Electricity Demand

	Model	RMSE	MAE
0	SARIMAX	2571.600148	1957.635746
1	Gradient Boosting	2948.594168	2212.911688
2	Random Forest	3042.194388	2238.531666
3	SARIMA	2042.323287	2613.567408
4	Seasonal Naive	2710.089546	3255.523670
5	Drift	3796.852789	4499.269103
6	Naive	3597.377895	4510.480290
7	Mean	4230.102079	4742.542591

Table 1. Comparison of Forecasting Model Performance (MAE and RMSE)

According to the performance measures, it is evident that the model named (write your best model here) performed with minimal forecasting errors, compared to the benchmarks. In general, the sophisticated models were able to predict the values more precisely than the simple benchmarks due to the presence of the trend, seasonality, and nonlinearity within the series.

6. Critical Analysis and Discussion

The results indicate that the advanced models outperformed the benchmark models because they were able to capture the seasonal and temporal aspects of the electricity consumption. The benchmark models had greater errors since they were unable to capture complex patterns in the time series data, while SARIMA and SARIMAX had superior performance due to modelling of seasonality and temperature effect.

The model choices had an effect on the outcome. The weekly aggregation minimized any noise, brought out the seasonality effects and the 80/20 chronological split ensured there was realistic testing on data which had not been seen before. AIC was used to help select parameters for the models.

Results from forecast comparisons revealed that the models generally followed the trend of demand but had difficulty when there were any abrupt changes or high-demand days. This implies that some more parameters such as holidays, renewable energy, and weather data can help increase accuracy. However, some of the major limitations are that there is only a limited period of data and fewer external parameters.

7. Conclusions and Future Improvements

This research compared various prediction models used for German energy demand, which include traditional models, SARIMA, SARIMAX, machine learning models, and LSTM. It is evident from the results that advanced models performed better due to seasonal patterns, temporal dependency, and exogenous factors like temperature. The most effective model was (insert your best model) since it had the lowest values of RMSE and MAE.

It can be seen from the forecasts that good prediction of electricity demand will help to make proper energy planning and decision making. It was found that there were some errors in the forecast when there is an unexpected change in the demand and during peak times.

Future work may also entail expanding the dataset by incorporating other external factors like holidays, energy production from renewables, electricity prices, and weather factors. Future work can also be done using more extensive historical data, ensemble forecasting techniques, and employing other machine learning models like transformers.

8. References

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